Momentum Continuation Screener

V2 Technical Specification & Strategy Ruleset

System Framework · Hard-Veto Gates · Multi-Timeframe Momentum · Implementation Spec

1 Strategy Scope & Pipeline Flow

The **Momentum Continuation Screener** evaluates End-of-Day (EOD) daily price bars to isolate high-conviction swing trading opportunities. The strategy focuses strictly on established macro uptrends experiencing temporary pullbacks to dynamic value areas.

Evaluation is executed through a **sequential multi-stage pipeline**. Failure at any hard gate immediately defines candidate status classification:

1. **Step 1 — Trend Gate:** Macro Baseline & MTF Momentum Hard-Veto Filter.
2. **Step 2 — Location Gate:** Pullback Proximity & Value Area Alignment.
3. **Step 3 — Oscillator Trigger Gate:** Short-term Momentum Turning Point Confirmation.

2 Step 1 — Trend Gate (Macro Baseline & MTF Momentum)

The Trend Gate validates structural trend health and higher-timeframe cyclical momentum. It functions as a hard-veto filter.

2.1 Layer 1: Structural Macro Trend (Daily Timeframe)

An asset must simultaneously satisfy three daily trend conditions:

Layer 1 Core Equation

$$\text{Rule: } \text{Close}_t > \text{SMA}_{200,t} \quad \text{AND} \quad \text{SMA}_{50,t} > \text{SMA}_{200,t} \quad \text{AND} \quad \text{SMA}_{50,t} > \text{SMA}_{50,t-20}$$

- **Secular Baseline:** Closing price must remain strictly above the 200-day Simple Moving Average ($\text{Close}_t > \text{SMA}_{200,t}$).
- **Alignment:** The 50-day SMA must sit above the 200-day SMA ($\text{SMA}_{50,t} > \text{SMA}_{200,t}$), maintaining a Golden Cross regime.
- **Expansion:** The slope of the 50-day SMA over a trailing 20-session window must be strictly positive ($\text{SMA}_{50,t} > \text{SMA}_{50,t-20}$).
- **Structural Flexibility:** Price is explicitly permitted to trade below the 50-day SMA during pullbacks, provided the 50-day SMA slope stays positive.

2.2 Layer 2: Multi-Timeframe (MTF) Cyclical Momentum

Layer 2 checks Stochastic momentum across Monthly (M) and Weekly (W) timeframes as a hard veto filter.

For any timeframe $TF \in \{\text{Monthly (M)}, \text{Weekly (W)}\}$:

$$\%K_{TF,t} = \textbf{SMA}_3\left(\frac{C_{TF,t} - L_{14,TF}}{H_{14,TF} - L_{14,TF}} \times 100\right), \quad \%D_{TF,t} = \textbf{SMA}_3\left(\%K_{TF,t}\right)$$

1. **Monthly Momentum Filter:** Requires that monthly $\%K$ is greater than $\%D$, or actively rising session-over-session:

$$(\%K_{M,t} > \%D_{M,t}) \vee (\%K_{M,t} - \%K_{M,t-1} > 0)$$

2. **Weekly Regime Filter:** Requires that weekly $\%K$ is rising session-over-session and at or above a baseline value of 20:
$$(\%K_{W,t} > \%K_{W,t-1}) \wedge (\%K_{W,t} \geq 20)$$

3 Step 2 — Location & Pullback Proximity Gate

Evaluates whether price has pulled back into a tight value area defined by moving average proximity and volatility-adjusted depth.

3.1 Pullback & Proximity Metrics

- **Recent Decline:** $\text{Close}_t < \text{Close}_{t-3}$
- **Pullback Depth:** $1.0 \leq \frac{\text{HighestHigh}_{20} - \text{Close}_t}{\text{ATR}_{14}} \leq 3.0$
- **EMA Proximity:** $-1.0 \leq \frac{\text{Close}_t - \text{EMA}_{20,t}}{\text{ATR}_{14}} \leq 0.5$

3.2 Dynamic EMA Reference & Grace Window Specifications

- **Dynamic Reference Line Selection (EMA_{ref}):** Selects EMA_8 or EMA_{21} based on proximity to the closing price:
$$\text{EMA}_{\text{ref}} = \begin{cases} \text{EMA}_8 & \text{if } |\text{Close}_t - \text{EMA}_8| \leq |\text{Close}_t - \text{EMA}_{21}| \\ \text{EMA}_{21} & \text{otherwise} \end{cases}$$
- **Proximity Threshold ($\tau_{\text{loc}} = 3.0\%$):** The percentage distance (D_{pct}) from the reference line must be $\leq 3.0\%$:
$$D_{\text{pct}} = \left| \frac{\text{Close}_t - \text{EMA}_{\text{ref}}}{\text{EMA}_{\text{ref}}} \right| \times 100 \leq 3.0\%$$
- **Grace Window Rule ($N = 5$ Sessions):** Location compliance ($\text{Location}_{\text{pass}}$) is satisfied if proximity was met on the current session, or during any of the prior 5 sessions:
$$\text{Location}_{\text{pass}} = (D_{\text{pct},t} \leq 3.0\%) \vee \left(\bigvee_{i=1}^{5} (D_{\text{pct},t-i} \leq 3.0\%) \right)$$

4 Step 3 — Oscillator Trigger Gate

Confirms short-term momentum turning points to verify pullback completion using Stochastic $(14, 3, 3)$.

4.1 Trigger Conditions

Trigger & Zone Constraints

$$\text{Crossover}_t = (\%K_t > \%D_t) \wedge (\%K_{t-1} \leq \%D_{t-1})$$
$$\text{Zone}_t = (\%K_t \leq 30) \vee (\%D_t \leq 30) \quad \text{(Overbought boundary: } \geq 70)$$

- **Grace Window Rule ($N = 5$ Sessions):** Trigger compliance ($\text{Oscillator}_{\text{pass}}$) is satisfied if an oversold crossover occurred on the current session, or within any of the prior 5 sessions:
$$\text{Oscillator}_{\text{pass}} = (\text{Crossover}_t \wedge \text{Zone}_t) \vee \left(\bigvee_{i=1}^{5} (\text{Crossover}_{t-i} \wedge \text{Zone}_{t-i}) \right)$$

5 Status Classification & UI Structure

The screener assigns each candidate stock to one of three primary status tiers:

Status Tier	Classification Requirements & Logic
Qualified	Meets all conditions: Passes Step 1 (Trend Gate), Step 2 (Location Gate, today or via 5-session grace window), and Step 3 (Oscillator Trigger Gate, today or via 5-session grace window).
Developing	Passes Step 1 (Trend Gate), but is actively awaiting pullback location alignment or an oscillator trigger signal (includes recent grace window tracking).
Other Monitored	Default watchlist or candidate setups failing the hard Trend Gate (Step 1).

6 Reference Implementation Pseudo-Code

Below is the python implementation snippet for Claude Code integration:

```python
def evaluate_momentum_continuation(symbol, data):
    # Step 1: Trend Gate (Layer 1 & Layer 2 Hard Veto)
    layer1_pass = (data.close[-1] > data.sma200[-1]) and \
                  (data.sma50[-1] > data.sma200[-1]) and \
                  (data.sma50[-1] > data.sma50[-20])

    monthly_rising = (data.stoch_k_monthly[-1] > data.stoch_d_monthly[-1]) or \
                     (data.stoch_k_monthly[-1] > data.stoch_k_monthly[-2])
    weekly_rising = (data.stoch_k_weekly[-1] > data.stoch_k_weekly[-2]) and \
                    (data.stoch_k_weekly[-1] >= 20)

    if not (layer1_pass and monthly_rising and weekly_rising):
        return {"status": "INELIGIBLE_TREND", "failed_step": 1}

    # Step 2: Location Gate
    highest_high_20 = max(data.high[-20:])
    atr_14 = data.atr14[-1]

    decline_pass = data.close[-1] < data.close[-4]
    depth_val = (highest_high_20 - data.close[-1]) / atr_14
    depth_pass = (1.0 <= depth_val <= 3.0)

    ema_val = (data.close[-1] - data.ema20[-1]) / atr_14
    ema_pass = (-1.0 <= ema_val <= 0.5)

    if not (decline_pass and depth_pass and ema_pass):
        return {"status": "INELIGIBLE_PULLBACK", "failed_step": 2}

    # Step 3: Oscillator Gate
    k_t, d_t = data.stoch_k[-1], data.stoch_d[-1]
    k_prev, d_prev = data.stoch_k[-2], data.stoch_d[-2]

    crossover = (k_t > d_t) and (k_prev <= d_prev)
    zone = (k_t <= 30) or (d_t <= 30)

    if not (crossover and zone):
        return {"status": "DEVELOPING", "failed_step": 3}

    return {"status": "EOD_QUALIFIED", "failed_step": None}
```